

Chain Reaction AI Report

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Overview

We built console AI for Chain Reaction game using minimax with alpha-beta. We test five heuristics and also comparison of search depth with time limits. This report show how good each heuristic and depth setting.

Heuristic Design and Rationale

We designed five heuristics targeting key strategic aspects of Chain Reaction:

H1: Orb Count Difference

Measures material advantage by counting total orb difference between players. While intuitive, this can mislead since tactical positioning often outweighs raw orb count in Chain Reaction.

H2: Close to Explosion

Calculates explosion readiness using orb-to-critical-mass ratios. Higher values indicate cells nearing explosion, capturing tactical preparation for chain reactions and positional threats.

H3: Corner/Edge Advantage

Counts strategic position control. Corners (critical mass = 2) and edges (critical mass = 3) explode easier than center cells (critical mass = 4), providing tactical advantages.

H4: Ready to Explode

Identifies cells requiring just one more orb to explode. These represent immediate tactical opportunities that can trigger game-changing cascades.

H5: Hybrid Evaluation

Combines all heuristics with optimized weights: orb difference (1.0), explosion readiness (3.5), edge advantage (2.0), ready-to-explode (4.0). Weights prioritize tactical positioning over material advantage, reflecting Chain Reaction's explosive nature where single moves can reverse game outcomes.

Experimental Setup

We use 9x6 board. Each matchup played 5 games. Two type of experiments:

- Heuristic vs Heuristic: depth=3, no time limit, 10 unique pairs of heuristics 1–5, each pair 5 games.
- Depth Comparison: depth-3 (no limit) vs depth-4 with time limit 5,4,3 seconds, each setting 5 games.

Heuristic Results

Figure 1 show win counts graph. Figure 2 show durations.

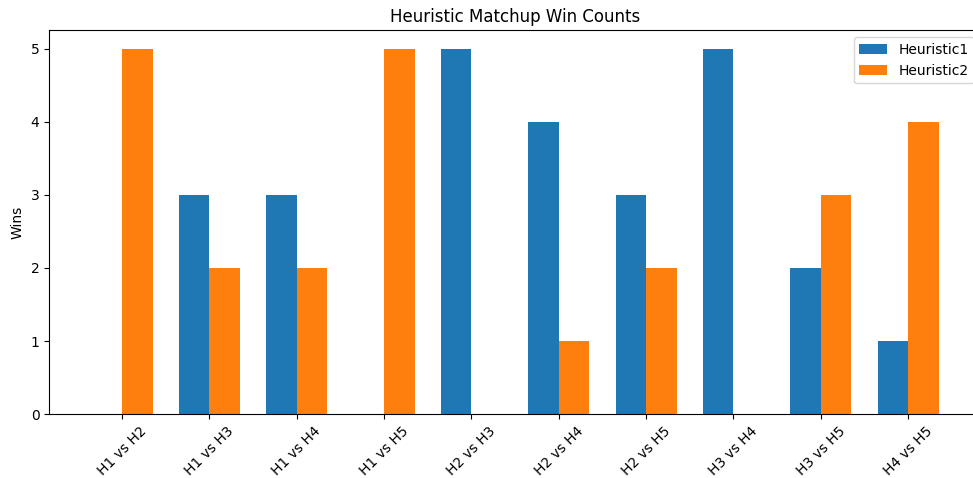


Figure 1: Win counts for heuristic matchups

Depth Results

Figure 4 show win counts for depth comparison. Figure 5 show average durations.

Discussion

From heuristic test, hybrid heuristic (H5) win more, it strong in both offence and defense. Simple heuristics like orb count (H1) lost many. H2 and H3 sometime good but not consistent.

Depth test show depth-4 only beat depth-3 if got enough time(8, 10s) which we can observe in fig-4 . If time is short (3s or 4s), depth-3 unlimited do better. Means must tradeoff depth and time.

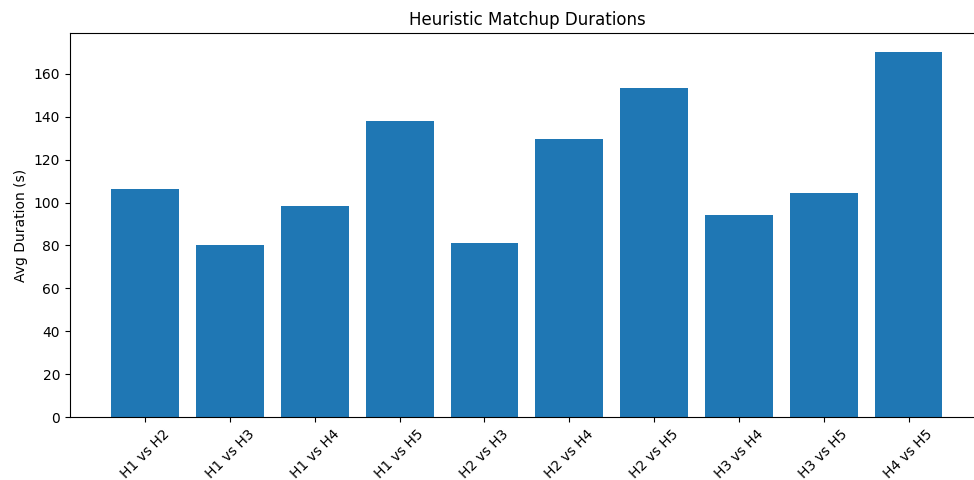


Figure 2: Avg duration for heuristic matchups

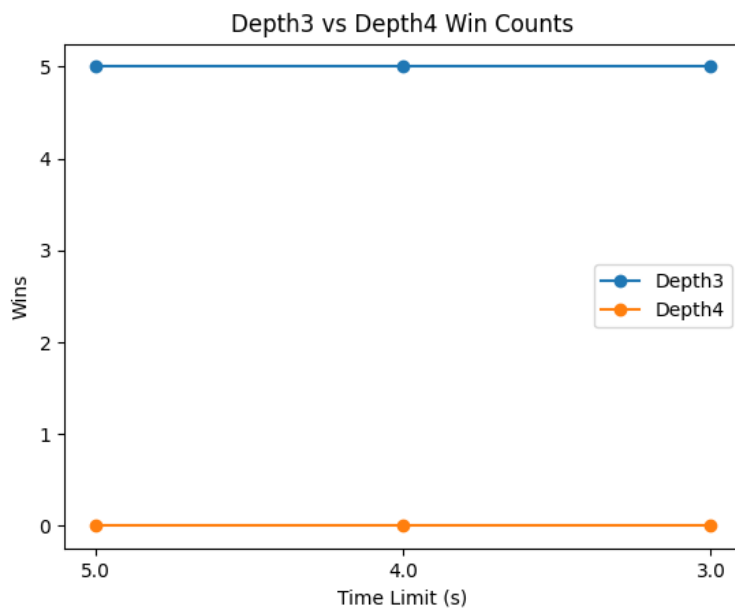


Figure 3: Wins: depth-3 vs depth-4 under time limits

Random play lose every time. this game need proper thinking and looking ahead. sometime a small move can create big explosion chain. if u don't think or plan, u just explode ur own orbs and help enemy. also if u don't care about corner or edge or enemy's threat, u easily fall into trap. so this game not work with random move, u must calculate and play smart.

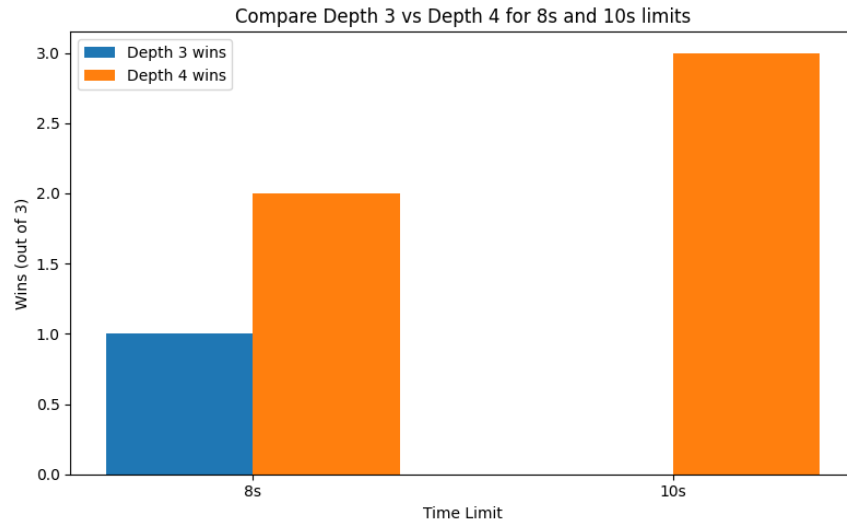


Figure 4: Wins: depth-3 vs depth-4 under time limits

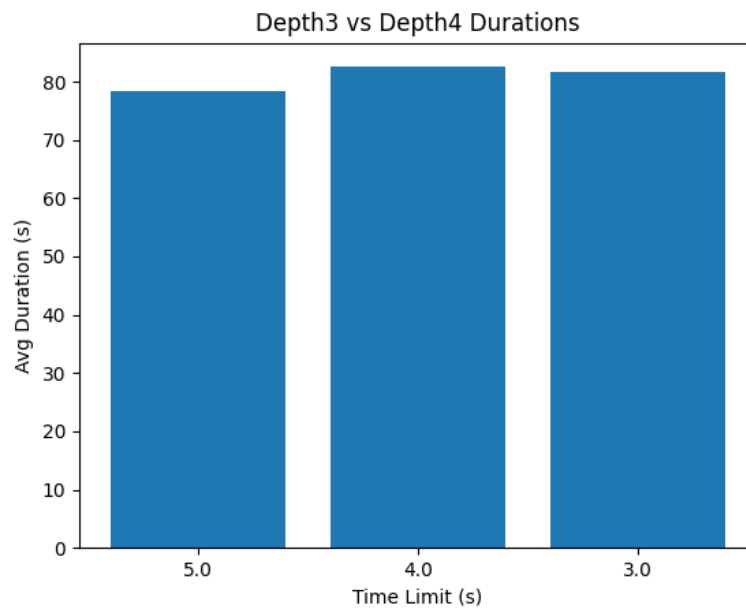


Figure 5: Avg duration by time limit

Conclusion

I done all tasks: console UI, minimax with alpha-beta, 5 heuristics, experiments. Hybrid heuristic is best at depth3. Depth4 require enough time to improve. This findings help choose right heuristic and search setting.